

Question:

Stephen placed an online order for groceries. His final bill came to \$40.00. Because this was through a delivery vendor, they tacked on a 25% fee to his final total and charged him \$3.00 in delivery fees. Stephen also added a \$4.00 tip. After the extra fees, what was the final price of Stephen's groceries?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Understand the structure of the bill .
- Stephen 's original grocery order cost is not given directly , but we know the final bill after fees and tip is \$ 40 . 00 .
- However , the problem says : " His final bill came to \$ 40 . 00 " â this is the total *after* delivery fees and tip ? Or before ?

Letâs read carefully :

> " His final bill came to \$ 40 . 00 . Because this was through a delivery vendor , they tacked on a 25 % fee to his final total and charged him \$ 3 . 00 in delivery fees . Stephen also added a \$ 4 . 00 tip . "

Wait â if the final bill is \$ 40 . 00 , and delivery fees and tip are *added on* , then the \$ 40 . 00 must be the base grocery price *before* fees and tip .

But the wording is ambiguous .

Letâs re-read :

> " Stephen placed an online order for groceries . His final bill came to \$ 40 . 00 . Because this was through a delivery vendor , they tacked on a 25 % fee to his final total and charged him \$ 3 . 00 in delivery fees . "

This suggests that the \$ 40 . 00 is the *total after* fees and delivery .

But then it says : â charged him \$ 3 . 00 in delivery feesâ â so delivery fee is \$ 3 . 00 â and â tacked on a 25 % fee to his final totalâ â this seems contradictory .

Wait â â tacked on a 25 % fee to his final totalâ â that would mean the final total is increased by 25 % â but then it says delivery fee is \$ 3 . 00 â so perhaps the \$ 40 is the total *after* all fees ?

But the phrasing is confusing .

Alternative interpretation :

Letâs suppose the \$ 40 is the *original grocery cost* .

Then :

- Delivery fee : \$ 3 . 00
- 25 % fee on the *final total* â so 25 % of the total after delivery and before tip ?

But then it says â Stephen also added a \$ 4 . 00 tip .â

So the total final price is :

Grocery cost + delivery fee + tip , with a 25 % fee on the total ?

Wait â the sentence says : â they tacked on a 25 % fee to his final total and charged him \$ 3 . 00 in delivery fees .â

So two fees :
- 25 % fee on the *final total* ?
- \$ 3 delivery fee ?

But that would mean the total is composed of :

Letâs suppose the grocery cost is X .

Then :

- Delivery fee : \$ 3 . 00
- 25 % fee on the *final total* â this is confusing because it says " tacked on a 25 % fee to his final total " â so perhaps the " final total " is the grocery cost plus delivery ?

Wait â perhaps the \$ 40 is the total *after* fees and tip ?

But then we have :

- Final bill = \$ 40 . 00
- Delivery fee = \$ 3 . 00
- Tip = \$ 4 . 00
- 25 % fee on the final total ?

That would mean the total is \$ 40 , which includes delivery and tip , but the 25 % fee is *on the final total* â so the grocery cost would be less .

But the 25 % fee is *tacked on to the final total* , which would make it even larger â so that doesnât make sense .

Alternative interpretation :

Maybe the \$ 40 is the grocery cost .

Then :

- Delivery fee : \$ 3 . 00
- 25 % fee on the grocery cost : 25 % of \$ 40 = \$ 10 . 00
- Tip : \$ 4 . 00